

Video-of-Thought: Step-by-Step Video Reasoning from Perception to Cognition

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Background: Video Understanding is Transforming the World

- Video powers safety, robotics, assistive tech, entertainment
- Decoding actions, causes, and outcomes is essential
- AI must link perception with reasoning for reliability

Outline

From Perception to Cognition: Why Video Reasoning Is Hard

- Beyond objects: track interactions, intentions, and causality
- Requires pixel-level grounding and commonsense fusion
- Human-like stepwise reasoning is a useful model

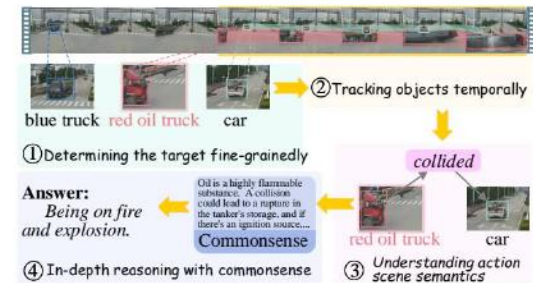


Figure: 类人视频推理：从像素定位/跟踪到动作场景语义的多步

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- Weak fine-grained spatiotemporal grounding
- Shallow cognition; hallucinated actions/intentions
- Text CoT alone ignores video dynamics and evidence

- How to ground objects and actions at pixel level?
- How to reason step-by-step over time reliably?
- How to verify answers with evidence and commonsense?

- MotionEpic: video MLLM with STSG grounding
- Video-of-Thought: five-step reasoning from pixels to cognition
- Grounding-aware tuning aligns video and STSGs

- STSG-centric MotionEpic for precise grounding
- VoT: targets → tracklets → actions → rank → verify
- SoTA across complex VideoQA; strong zero-shot

- ViT-L/14 encoder + Q-Former projector
- Vicuna-7B LLM for language and control
- Recurrent Graph Transformer for multi-frame STSGs

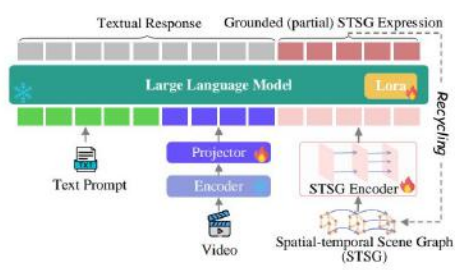


Figure: Overview of the MotionEpic video MLLM.

- Nodes: objects with category, bbox, embeddings
- Edges: relations; temporal links form tracklets
- Generated STSGs become grounded rationales

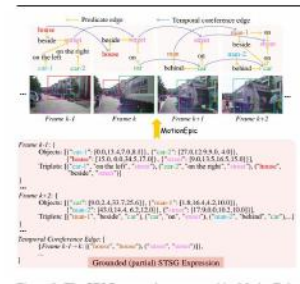


Figure: MotionEpic生成的STSG表达及对应STSG结构示意图

- L1: video–STSG pair verification
- L2: full STSG generation from video
- L3: tracklets from action text queries
- L4: action description + tracklets from key objects
- L5: bbox → label + tracklet mapping

- Chain steps from perception to cognition
- Reuse grounded STSGs across reasoning steps
- Supports STSG-free inference after tuning



Figure: VoT framework overview; I/O and prompts in Appendix.

VoT Step 1: Target Identification

- Extract key entities and actions from question
- List targets and temporal scopes
- Prime subsequent grounding requests

VoT Step 2: STSG-Based Tracking

- Generate partial STSG tracklets for targets
- Enforce temporal coreference and boxes
- Emit evidence snippets for reuse

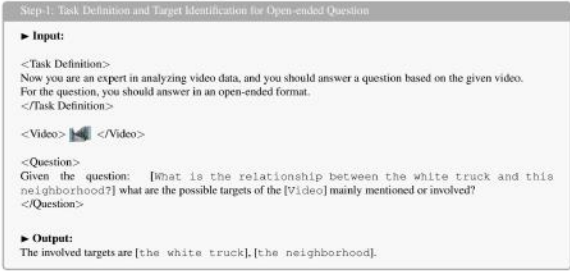


Figure: ▷ Step-2: The detailed prompt for Step-2 is shown as follows:

VoT Step 3: Action Analysis with Commonsense

- Fuse tracklets with neighbor STSG context
- Leverage LLM knowledge for intent/causality
- Produce candidate action hypotheses

VoT Step 4: Answer Ranking

- Score candidates by evidence coverage
- Prefer parsimonious, temporally coherent answers
- Handle open-ended via implicit multi-choice

- Grounding check: pixel/tracklet consistency
- Commonsense check: plausibility and causality
- Re-rank or abstain on contradictions

- Datasets: VLEP, STAR, IntentQA, Social-IQ, Causal-VidQA, NEX-T-QA
- Training: WebVid pretrain; instruction tuning corpora
- Baselines: Video-LLaVA, Video-ChatGPT, SeViLA, VLAP, others

Key Results: Broad VideoQA Gains (Table 1)

- VoT with MotionEpic outperforms prior SoTA
- Largest lifts on complex reasoning subsets
- Structured steps beat vanilla CoT prompting

Table: Table 1: Results on four VideoQA datasets. STAR data includes four subsets: Interaction (Int.), Sequence (Seq.), Prediction (Pre.), Feasibility (Fea.). The best scores of baselines are underlined, and the new best results are bold.

Model	VLEP	STAR	Int. Seq. Pre. Fea.				IntentQA	Social-IQ	2-Way 4-Way	
			Int.	Seq.	Pre.	Fea.			2-Way	4-Way
• SoTA baselines										
InternVideo	63.9		62.7	65.6	54.9	51.9	-	-	-	-
LLaMA-VQA	71.0		66.2	67.9	57.2	52.7	-	-	-	-
VLAP	69.6		70.0	70.4	65.9	62.2	-	-	-	-
SeViLA	68.9		63.7	70.4	63.1	62.4	-	-	-	-
VideoChat	62.0		63.2	66.8	54.1	49.6	59.3	67.7	37.8	
Video-LLaVA	65.8		64.3	67.0	56.5	50.1	62.5	68.9	39.2	
• CoT										
Video-LLaVA	65.7		65.0	67.7	57.8	52.0	63.2	69.5	40.4	
Video-LLaVA+STSG	67.0		65.9	68.9	58.7	53.7	64.9	70.4	41.7	
MotionEpic	68.2		66.8	69.6	60.6	57.4	66.1	71.7	43.0	
• VoT										
MotionEpic	73.4		71.5	72.6	66.6	62.7	70.8	72.8	45.0	

Causal-VidQA: Description, Explanation, Prediction, Counterfactual

- Strongest on Explanation and Counterfactual
- Prediction A/R/AR improved notably
- VoT consistently surpasses CoT variants

Table: Table 2: Results on Causal-VidQA data. D: Description, E: Explanation, P: Prediction, C: Counterfactual.

Model	Acc@D	Acc@E		Acc@P		AR					Acc@C		AR
				A	R	AR					A	R	AR
• SoTA baselines													
TranSTR	73.6	75.8				65.1	65.0	48.9	68.6	65.3	50.3		
Video-LLaMA	69.2	71.0				63.6	62.4	44.4	65.4	60.1	45.0		
VideoChat	72.9	73.9				65.2	63.1	45.9	66.0	62.7	45.8		
Video-ChatGPT	73.1	75.1				66.0	63.9	46.0	67.8	63.6	50.0		
Video-LLaVA	73.7	74.4				67.6	65.4	47.7	68.0	64.9	51.5		
• CoT													
Video-LLaVA	74.2	74.8				68.0	65.7	48.1	70.3	65.7	52.9		
Video-LLaVA+STSG	75.7	75.9				68.9	67.2	50.0	70.7	67.2	53.6		
MotionEpic	78.5	77.2				70.1	70.8	52.4	71.2	69.1	55.0		
• VoT													
MotionEpic	81.2	83.0				74.3	73.7	54.7	74.5	73.8	58.6		

- New best Acc@All, strong on temporal subset
- VoT $\geq$ CoT even with same backbone
- Evidence reuse improves causal answers

Table: Table 3: Results on NExT-QA data.

Model	Acc@All	Acc@C	Acc@T	Acc@D
• SoTA baselines				
InternVideo	63.2	62.5	58.5	75.8
HiTeA	63.1	62.4	58.3	75.6
LLaMA-VQA	72.0	72.7	69.2	75.8
SeViLA	73.8	73.8	67.0	81.8
VLAP	75.5	74.9	72.3	82.1
Video-LLaMA	60.6	59.2	57.4	72.3
VideoChat	61.8	62.5	61.5	74.6

- AVG 62.3% vs best baseline 56.0%
- w/o grounding verification: -3.3 points
- w/o commonsense verification: -2.2 points

Probing MotionEpic: Grounding Reliability

- Competitive with SoTA STSG parsers
- Near-human on objects, relations, actions
- Trustworthy perceptual evidence for VoT

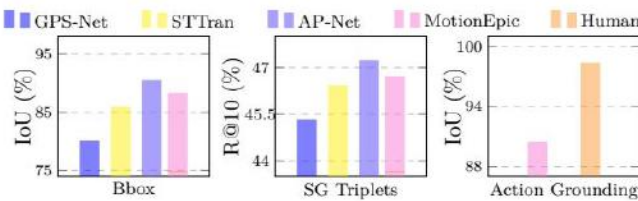


Figure: MotionEpic在物体/动作定位与场景图三元组分类表现

Ablation: Which Tuning Objectives Matter?

- Removing L2 or L4 hurts performance most
- STSG generation and action-track links are critical
- Validates design of multi-level objectives

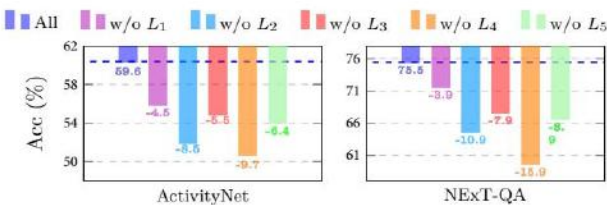


Figure: MotionEpic zero-shot drop after grounding-aware ablations

- Approaches human accuracy on hard subsets
- Fewer action semantics and commonsense errors
- VoT reduces hallucinations vs CoT baselines

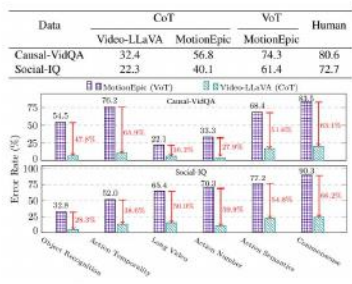


Figure: 上表：视频问答人工评估；下图：各类别错误率

- Accurate counting and temporal localization
- Evidence-backed tracklets shown clearly
- Baselines miss subtle temporal cues



Figure: 感知层级推理定性示例：正确以绿勾标注，错误以红叉标注。

- Fuse cultural/context knowledge with evidence
- Justified intentions and outcomes
- Stepwise VoT explains final answers



Figure: Qualitative examples of cognitive-level reasoning.

- Training cost and inference latency
- STSG parsing errors can propagate
- Potential misuse; require safeguards and licensing

- VoT: first effective CoT-style video reasoning
- MotionEpic: STSG-grounded, explainable perception
- New SoTAs and strong zero-shot generalization

- Extend VoT to retrieval, captioning, forecasting
- Scale backbones and external knowledge sources
- Real-time STSG parsing; richer counterfactuals

- Thank you for your attention!
- Questions and feedback are welcome.

- Thanks to collaborators and supporting institutions
- Appreciation for dataset and toolkit contributors